

Joshua O'Callaghan

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EXPERIENCE

Software Engineer - Cloud

June 2023 - Nov 2024

Intel

- Optimized Kafka workload for client, doubling throughput while reducing CPU utilization by 20%
- Improved compression workloads for client, achieving a 12x increase in compression pipeline speed
- Managed cloud environments (AWS, Azure, GCP) and internal server systems for the broader team
- Ensured all systems were vulnerability-free while reducing cloud costs by 6% year over year

Software Engineer - Media Processing

May 2021 - June 2023

Intel

Chandler, AZ

- Collected FFmpeg data and performed media analysis across various cloud/physical machines
- Created online tool for visualizing and maintaining performance numbers across Intel product line
- Automated benchmarking and data collections to improve efficiency by 500%
- Led a team of engineers and stakeholders to deliver competitive analysis benchmarking software

Software Engineer

Sept 2019 - May 2021

Intel

Chandler, AZ

- Visualized data for error analysis with various visualization tools such as Plotly, PowerBI and Tableau
- Assembled various Python and Bash scripts for machine fab automation

Software Engineering Intern

June 2019 - Aug 2019

American Express

Scottsdale, AZ

- Redesigned data infrastructure for Platform Engineering team
- Administered 4 server nodes running unique applications and tools on a Kubernetes cluster

Software Engineering Teaching Assistant

Jan 2019 - June 2019

Arizona State University (CSE 340)

Tempe, AZ

- Guided students in formal language theory, covering regular expressions and context-free grammars
- Collaborated with the course instructor to refine teaching strategies and ensure effective delivery of content

SKILLS

Languages Proficient Python (6yrs) Intermediate SQL (3yrs) Java (2yrs) C/C++ (2yrs) Beginner Javascript C#

Software AWS, GCP, Azure, Terraform, Ansible, Docker, Kubernetes, Elasticsearch, Git, Bash

Soft Skills Leader, Communicative, Problem Solver, Innovator, Detail Oriented, Always Learning

PROJECTS

Disciple Assistant | Personal Project (~80 hours)

Dec 2024

- Used JS, YT pub sub, YT API, ChatGPT API and SQL to deliver discipleship automation for Impact Church

Weather Dashboard | Personal Project (~50 hours)

Oct 2024

- Used Python, Elasticsearch, Logstash, Kibana and OpenWeather API to deliver an automated real-time dashboard

Discord Bot Development | Personal Project (~30 hours)

Mar 2024

- Used Python, Discord API and ChatGPT API to create a management bot for a discord server

EDUCATION

Arizona State University, Ira Fulton School of Engineering

Dec 2020

Master of Computer Science

Arizona State University, Ira Fulton School of Engineering

Dec 2019

Bachelors of Science in Computer Science

Excelling Coursework: Data Structures and Algorithms, Advanced Algorithms, Distributed Databases

ACTIVITIES

Machine Learning Data Engineer - Evaluated coding performance on Claude, ChatGPT, Gemini, and other LLMs July 2024

Google HashCode Competition - Placed 44th in US, 844th globally

Feb 2020